Review Test Submission: Assignment Week 3

User	Mr W (Wouter) W Nel
Course	EBN 122 S2 2017
Test	Assignment Week 3
Started	8/5/17 5:01 PM
Submitted	8/5/17 7:21 PM
Due Date	8/7/17 11:59 PM
Status	Completed
Attempt Score	14.9 out of 20 points
Time Elapsed	2 hours, 20 minutes

Instructions

Read over the document *ClickUP Assignments Instructions*.

Please direct any enquiries regarding ClickUP assignments to EBN2017.ClickUptest@gmail.com

Save the answer to each question as you complete it.

At the end of the assignment click Save and Submit.

Give your final correct answer 3 significant digits.

Make sure you use the FULL STOP for your decimal place marker.

You must indicate the units, unless otherwise specified:

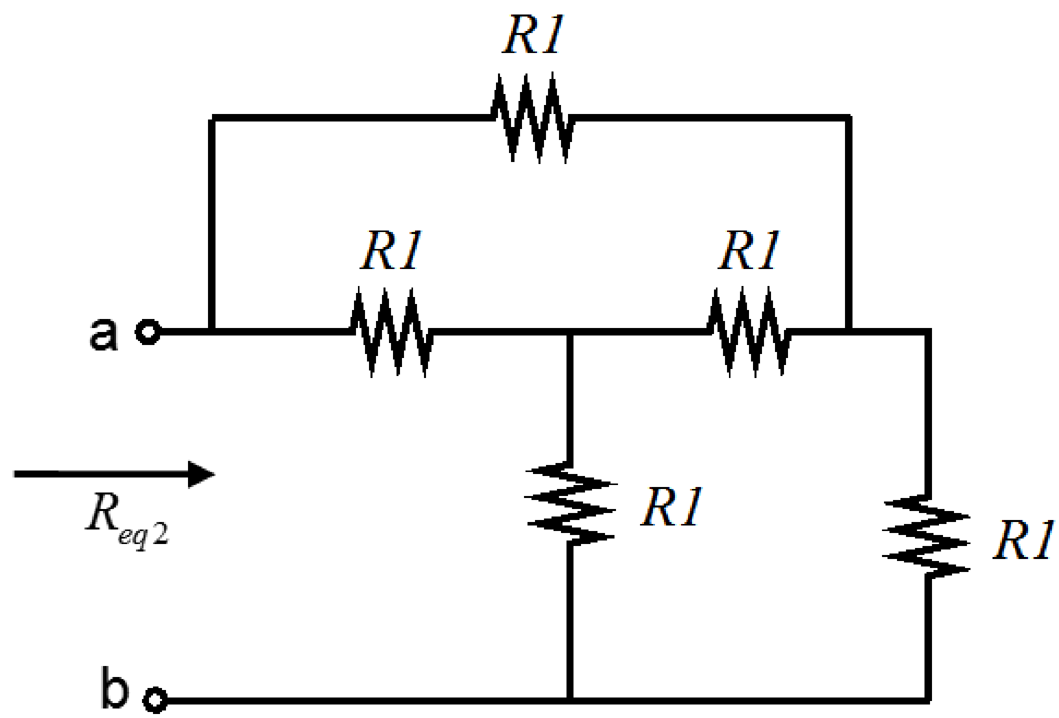
- There must be NO space between the answer and the units
- There must be between 1 and 3 digits to the left of the decimal point. *This means that you need enter 1kV rather than 1000V and 100mV rather than 0.1V*
- Write out 'ohm' to indicate Ω (use sigular)
- Use '**u**' to indicate micro.
- Only use pico(**p**), nano(**n**), micro(**u**), milli(**m**), kilo(**k**), mega(**M**), giga(**G)** and tera(**T**); do not use centi, deci and deca

Results Displayed	All Answers, Submitted Answers, Correct Answers, Feedback, Incorrectly Answered Questions

Question 1

2 out of 2 points

Calculate R_{eq2} when $R_1 = 21\Omega$.



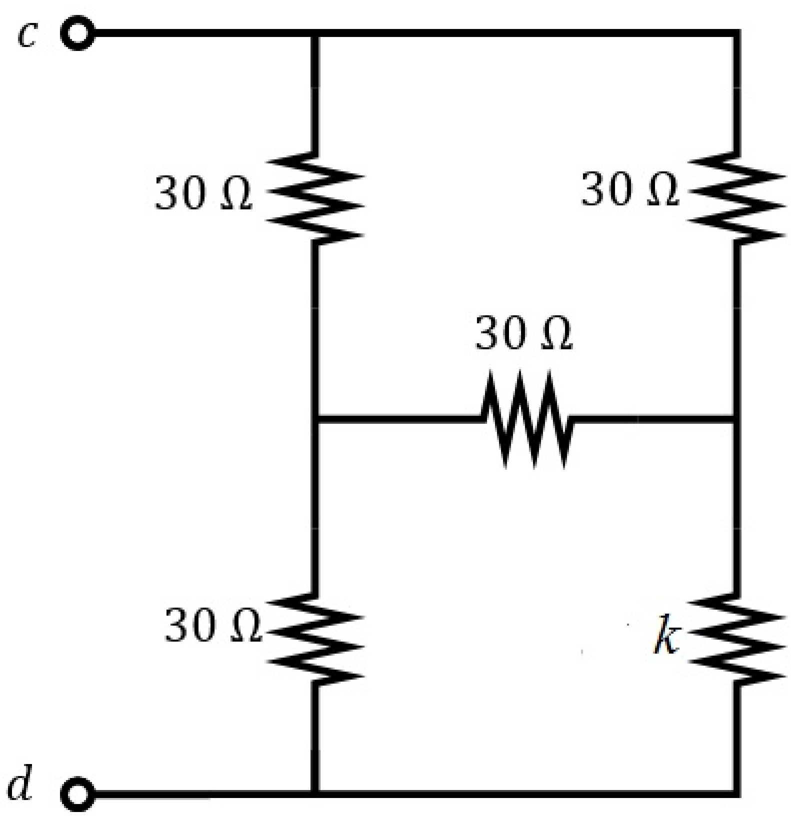
Selected Answer: 21ohm

Correct Answer: 21.00 ± 2% (ohm)

Question 2

2 out of 2 points

With k = 43Ω find R_{cd}



Selected Answer: 32.8ohm

Correct Answer: 32.80 ± 5% (ohm)

Question 3

2 out of 2 points

Determine α if I_1= 2.5 A

$$\begin{bmatrix} 4 & \alpha \\ -8 & 2 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 6 \\ -4 \end{bmatrix}$$

Selected Answer: -0.5

Correct Answer: -0.500 ± 5%

Question 4

2 out of 2 points

If I = 16 A, find m.

$$\begin{bmatrix} 5 & 8 \\ -1 & m \end{bmatrix} \begin{bmatrix} I \\ V \end{bmatrix} = \begin{bmatrix} 3 \\ -4 \end{bmatrix}$$

Selected Answer: -1.25

Correct Answer: -1.25 ± 5%

Question 5

2 out of 2 points

Determine V1, given k = 3:

$$\begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} k \\ -5 \end{bmatrix}$$

Selected Answer: 1V

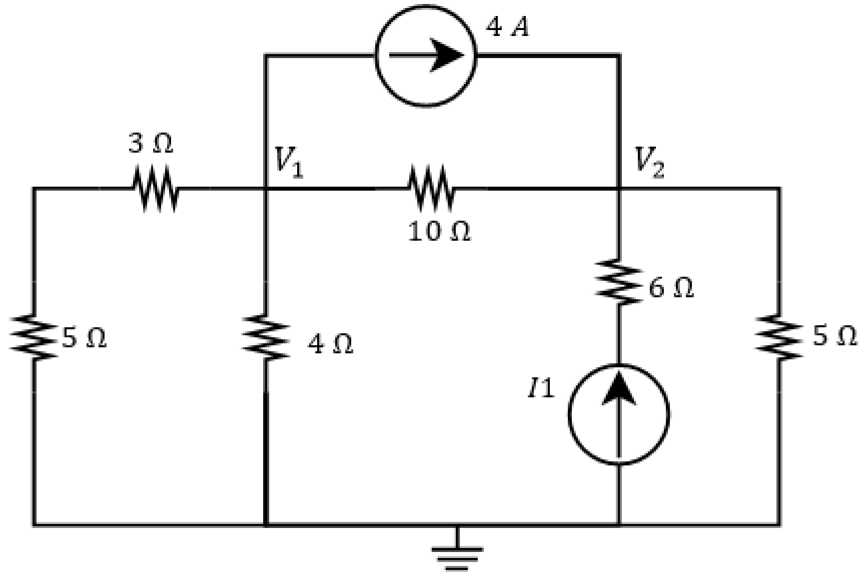
Correct Answer: 1.00 ± 2%

Question 6

2 out of 2 points

Use nodal analysis to set up two equations of the following form:
Provide h, if I1 = 4 A.

$$eV_1 + fV_2 = -160$$
$$-V_1 + gV_2 = h$$



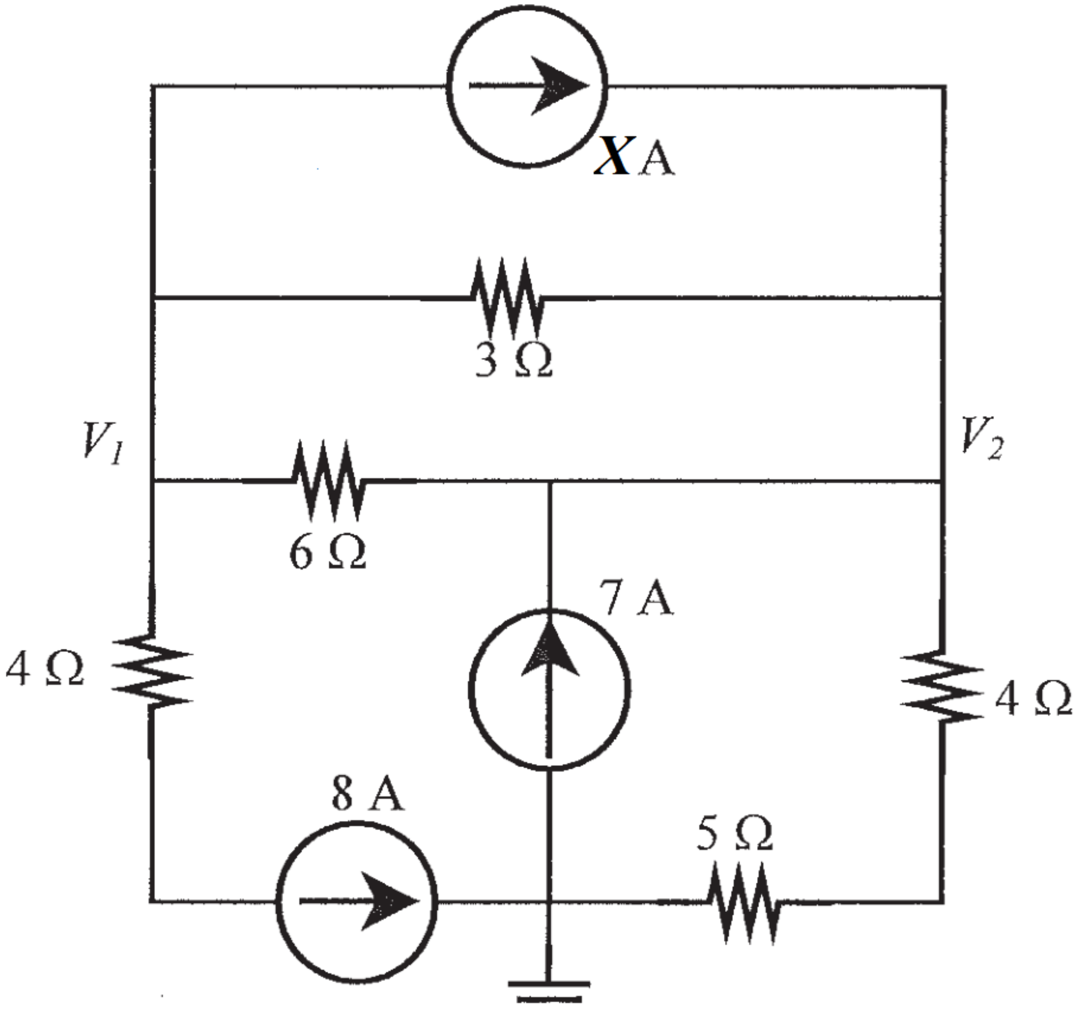
Selected Answer: 80

Correct Answer: 80.00 ± 5%

Question 7

0 out of 2 points

Given X = 7 A, use nodal analysis to set up an equation of the following form the for the circuit. Give the value of b:
$-9V_1 + 11V_2 = b$



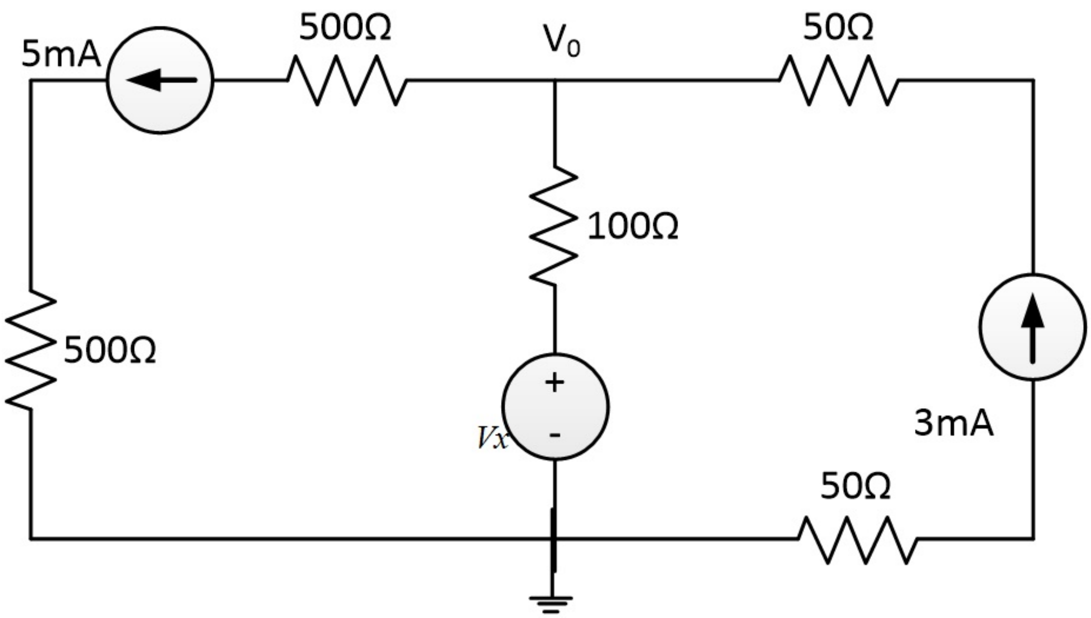
Selected Answer: -252

Correct Answer: 252.00 ± 5%

Question 8

2 out of 2 points

If V_x = 9 V find V_o.



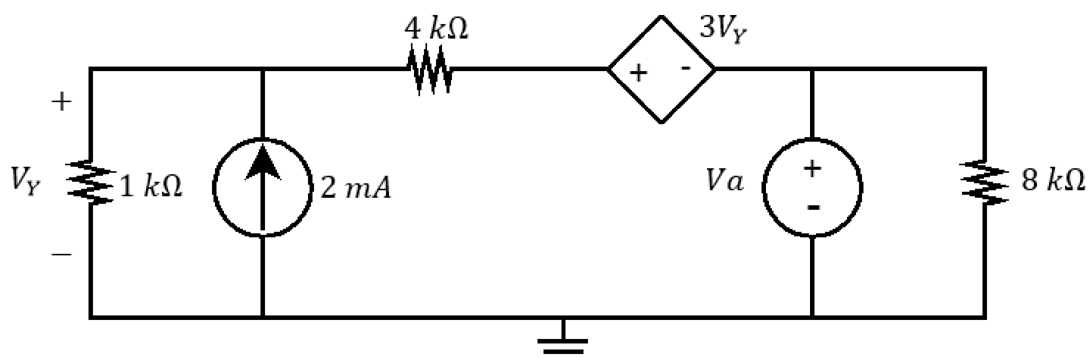
Selected Answer: 8.8V

Correct Answer: 8.80 ± 2% (V)

Question 9

0.4 out of 2 points

Use nodal analysis to determine V_y, when V_a = 5 V.



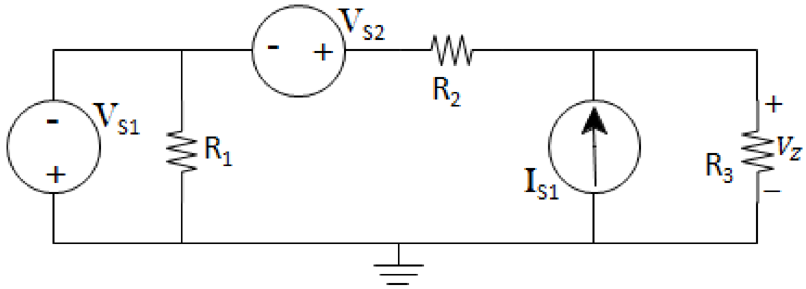
Selected Answer: 1.63V

Correct Answer: 6.50 ± 5% (V)

Question 10

0.5 out of 2 points

Determine the value of V_Z.
V_{S1}=7V, V_{S2}=19V, I_{S1}=33mA, R_1=82Ω, R_2=30Ω, R_3=66Ω



Selected Answer: 5.49V

Correct Answer: 8.931 ± 2% (V)

Saturday, August 5, 2017 7:22:02 PM SAST

← **OK**